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Review

Spatially resolved transcriptomics: a comprehensive review of their technological advances, applications, and challenges

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ABSTRACT

The ability to explore life kingdoms is largely driven by innovations and breakthroughs in technology, from the invention of the microscope 350 years ago to the recent emergence of single-cell sequencing, by which the scientific community has been able to visualize life at an unprecedented resolution. Most recently, the Spatially Resolved Transcriptomics (SRT) technologies have filled the gap in probing the spatial or even three-dimensional organization of the molecular foundation behind the molecular mysteries of life, including the origin of different cellular populations developed from totipotent cells and human diseases. In this review, we introduce recent progresses and challenges on SRT from the perspectives of technologies and bioinformatic tools, as well as the representative SRT applications. With the currently fast-moving progress of the SRT technologies and promising results from early adopted research projects, we can foresee the bright future of such new tools in understanding life at the most profound analytical level.

Introduction

The study of tissues within their own two-dimensional (2D) or three-dimensional (3D) setting is referred to as ‘spatial biology’, which has emerged as—and now represents—the latest frontier of molecular biology (Crosetto et al., 2015). Considering its immense importance, explosive popularity, and potential to offer future insight into currently pending research questions, the spatially resolved transcriptomics (SRT) technology was declared as the method of the year 2020 by the journal ‘*Nature Methods*’ (2021). Spatial data can be interpreted at the cellular and molecular levels in a way similar to how ‘Global Positioning System’ records location coordinates within a region to produce a map and track people or objects. Of note, SRT methods produce data and enable us to identify specific functional regions at the genome-wide scale by deciphering the variety of gene expression (Tian et al., 2022), to resolve cell type heterogeneity and intercellular communications by analyzing intrinsic expression features of genes and their physical proximity (Grisanti Canozo et al., 2022; Moffitt et al., 2022), and to even analyze RNA expression at the subcellular level (Cassella and Ephrussi, 2022). The power of SRT technologies in deciphering tissue complexity has helped us to provide atlases of critical biological processes, such as the development of tissues or even organoids (He et al., 2022; Lohoff et al., 2022), as well as human disease pathogenesis (Parigi et al., 2022; Cross et al., 2023).

From the first reported method of achieving molecular detection in tissues by *in situ* hybridization in 1969 (Gall and Pardue, 1969) to the recent emergence of various SRT technologies based on next-generation sequencing (NGS) (Ståhl et al., 2016; Chen et al., 2017; Rodriques et al., 2019; Vickovic et al., 2019; Liu et al., 2020c; Cho et al., 2021), SRT technology has ushered in revolutionary development due to the recent progress in sequencing technology and single-cell ‘-omics’. Generally, SRT technologies can be classified into two categories: (a) imaging-based and (b) sequencing-based methods, which are continuously optimized by mainly improving resolution and the capture capability (including the gene throughput

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Table 1
The main four aspects of SRT technologies based on sequencing and imaging strategies.

Strategy	Methods	Name	Captured	Sample type	References
Imaging-based	*In situ* hybridization based	smFISH	Targeted RNA	Cells	Femino et al., 1998
		RNAscope	Targeted RNA	Cells, PFA and PPFE	Wang et al., 2012
		seqFISH	Targeted RNA	Cells	Lubeck and Cai, 2012
		MERFISH	Targeted RNA	Cells, Tissue sections	Moffitt et al., 2016
		RNA SPOTs	Targeted RNA	Cells	Eng et al., 2017
		MERFISH + ExM	Targeted RNA	Cells, Tissue sections	Wang et al., 2018a
		osmFISH	Targeted RNA	Tissue sections	Codeluppi et al., 2018
		ClampFISH	Targeted RNA	Cells, Tissue sections	Rouhanifard et al., 2018
		seqFISH+	Targeted RNA	Cells, Tissue sections	Eng et al., 2019
		DNA Microscopy	Individual DNA or RNA molecules	Cells	Weinstein et al., 2019
		EASI-FISH	Targeted RNA	Tissue sections	Wang et al., 2021
		EEL FISH	Targeted RNA	Tissue sections	Borm et al., 2023
		CAD-HCR	Targeted RNA and antibody-based protein	Cells	Liu et al., 2022c
		MOSAICA	Targeted RNA and antibody-based protein	Cells, FFPE	Vu et al., 2022
		WTA	Targeted RNA	Cells	Zimmerman et al., 2022
		par-seqFISH	Targeted RNA	Microbial Cells	Dar et al., 2021
Sequencing-based	Micro-dissection based	tomo-seq	Transcriptome-wide	Fresh-frozen	Junker et al., 2014
		Geo-seq	Transcriptome-wide	Fresh-frozen	Chen et al., 2017
		GaST-seq	Transcriptome-wide	Plant	Giolai et al., 2019
		DSP	Transcriptome-wide and antibody-based protein	Fresh-frozen and FFPE	Merritt et al., 2020
		ZipSeq	Transcriptome-wide	Fresh-frozen	Hu et al., 2020
		PIC	Transcriptome-wide	Fresh-frozen	Honda et al., 2021
		immuno-LCM-RNAseq	Transcriptome-wide	Fresh-frozen	Zhang et al., 2022
		PuTi-spots	Transcriptome-wide	FFPE	Matsunaga et al., 2022
		Image-seq	Transcriptome-wide	Cells	Haase et al., 2022
	In situ sequencing-based	ISS	Targeted RNA	Cells, Tissue sections	Ke et al., 2013
		FISSEQ	Untargeted RNA	Cells, Tissue sections	Lee et al., 2014
		STARmap	Targeted RNA	Tissue sections	Wang et al., 2018b
		BARseq	Targeted RNA/ barcoding RNA	Cells, Tissue sections	Chen et al., 2019
		HybISS	Targeted RNA	Tissue sections	Gyllborg et al., 2020
		ExSeq	Untargeted/Targeted RNA	Cells, Tissue sections	Alon et al., 2021
		BOLORAMIS	Targeted RNA	Cells, Tissue sections	Liu et al., 2021
		BARseq2	Targeted RNA/ barcoding RNA	Cells, Tissue sections	Sun et al., 2021
		STARmap PLUS	Targeted RNA and antibody-based protein	Tissue sections	Zeng et al., 2023
		TEMPOmap	Nascent transcriptomes	Cells	Ren et al., 2022
		RIBOmap	Ribosome-bound mRNAs	Cells, Tissue sections	Zeng et al., 2022
		IISS	Targeted RNA	Fresh-frozen and FFPE	Tang et al., 2022
		SPRINTseq	Targeted RNA	Tissue sections	Chang et al., 2022
	In situ spatial barcoding based	ST	A-tailed RNA transcripts	Fresh-frozen	Ståhl et al., 2016
		Visium	A-tailed RNA transcripts and targeted genes	Fresh-frozen and FFPE	10xgenomics.com
		MASC-seq	A-tailed RNA transcripts	Single-cell	Vickovic et al., 2016
		Slide-seq	A-tailed RNA transcripts	Fresh-frozen	Rodriques et al., 2019
		HDST	A-tailed RNA transcripts	Fresh-frozen	Vickovic et al., 2019
		DBiT-seq (FFPE)	A-tailed RNA transcripts	FFPE	Liu et al., 2020a
		DBiT-seq	A-tailed RNA transcripts and antibody-based protein	Fresh-frozen	Liu et al., 2020b

Table 1 (*continued*)

Strategy	Methods	Name	Captured	Sample type	References
		Slide-seq/V2	A-tailed RNA transcripts	Fresh-frozen	Stickels et al., 2021
		XYZeq	A-tailed RNA transcripts	Fresh-frozen	Lee et al., 2021
		Seq-Scope	A-tailed RNA transcripts	Fresh-frozen	Cho et al., 2021
		sci-Space	A-tailed RNA transcripts	Fresh-frozen	Srivatsan et al., 2021
		Space-TREX	A-tailed RNA transcripts and CloneIDs	Fresh-frozen	Ratz et al., 2021
		SM-Omics	A-tailed RNA transcripts and antibody-based protein	Fresh-frozen	Vickovic et al., 2022
		Perturb-map	A-tailed RNA transcripts and CRISRP targeted genes	Fresh-frozen	Dhainaut et al., 2022
		Spatial-ATAC-RNA-seq	A-tailed RNA transcripts and Transposase chromatin	Fresh-frozen	Jiang et al., 2023
		spatial-CITE-seq	A-tailed RNA transcripts and antibody-based protein	Fresh-frozen	Liu et al., 2022d
		Stereo-seq	A-tailed RNA transcripts	Fresh-frozen	Chen et al., 2022a
		SPOTS	A-tailed RNA transcripts and antibody-based protein	Fresh-frozen	Ben-Chetrit et al., 2022
		SmT	Host transcriptome- and microbiome-wide	Fresh-frozen	Saarenpää et al., 2022
		Matrix-seq	A-tailed RNA transcripts	Fresh-frozen	Zhao et al., 2022
		xDbit	A-tailed RNA transcripts	Fresh-frozen	Wirth et al., 2022
		CBSST-Seq	A-tailed RNA transcripts	Fresh-frozen	Jin et al., 2022
		slide-TCR-seq	A-tailed RNA transcripts and targeted T cell receptor genes	Fresh-frozen	Liu et al., 2022b
		Ex-ST	A-tailed RNA transcripts	Fresh-frozen	Fan et al., 2022
		STRS	Total RNA	Fresh-frozen	McKellar et al., 2022
		Pixel-seq	A-tailed RNA transcripts	Fresh-frozen	Fu et al., 2022
		Spatial VDJ	Full-length T and B cell antigen receptor genes	Fresh-frozen	Engblom et al., 2022
		STOmics-GenX	A-tailed RNA transcripts without CRISPRclean reads	Fresh-frozen	Currenti et al., 2022

and capture efficiency). Given that all these methods are different in terms of workflow and data performance, we herein first provide a comprehensive overview of the current SRT-related technologies (Table 1) and summarize the technical advantages, drawbacks, and key processes toward the landmark techniques (Fig. 1), as well as future trends. We also review the standard SRT data analysis process, common software toolkits, and discuss in detail some of the key steps, including image registration and cell segmentation, single-cell resolution analysis, gene imputation, cell communication, as well as some of the challenges. Finally, we briefly describe the applications of SRT technologies in a wide spectrum of both basic research and translational medicine settings.

The revelation of spatially resolved transcriptomics technologies

Imaging-based spatially resolved transcriptomics

The imaging-based SRT technologies achieve spatial visualization by combining the chromogenic group with the probes, thus offering the natural advantage of reflecting the distribution of the target genes directly *in situ* (Langer-Safer et al., 1982). The major challenges of this method include (a) improving the signal intensity and (b) increasing the number of gene detection. Historically, radioactive elements were first used to visualize transcripts in 1973 (Harrison et al., 1973), with continuous innovation hallmarked by later emerged chemical chromogenic groups (Tautz and Pfeifle, 1989), fluorescence-modified groups (Singer and Ward, 1982), and, until recently, the widely used fluorescence-modified single-base method (Femino et al., 1998). To enhance the gene detection sensitivity, several signal amplification methods are developed. The examples include padlock nucleotide modification with *in situ* rolling circle amplification (Larsson et al., 2010), branched DNA binding (Battich et al., 2013), and multiplex smFISH (Lubeck and Cai, 2012). The most innovative and widely used imaging-based methods recently are sequential fluorescence *in situ* hybridization (seqFISH) (Lohoff et al., 2022) and multiplexed error-robust fluorescence *in situ* hybridization (MERFISH) (Chen et al., 2015b). Among several similar approaches, seqFISH and MERFISH have been established to address the needs of genome-wide approaches and to overcome the limitations of single-molecule fluorescence *in situ* hybridization, in which every single spot corresponds to a single transcript (Burgess, 2019; Lohoff et al., 2022). More specifically, in seqFISH, the same probes undergo distinct rounds of hybridization based on distinct fluorophores on each round, thus creating a gene-specific association of the types and rounds of fluorophores (Lubeck et al., 2014). Another technique, seqFISH+ associates initial gene-specific hybridization with several binding sites for sequential rounds of hybridization (Eng et al., 2019). MERFISH represents an adjustment to the above methods, by reducing the number of sequential orders of fluorophores and applying several binding sites (Chen et al., 2015b). Of note, the approach of replacing a high number of fluorophores by a high number of binding sites leads MERFISH to be less error-prone during the exponential increase in

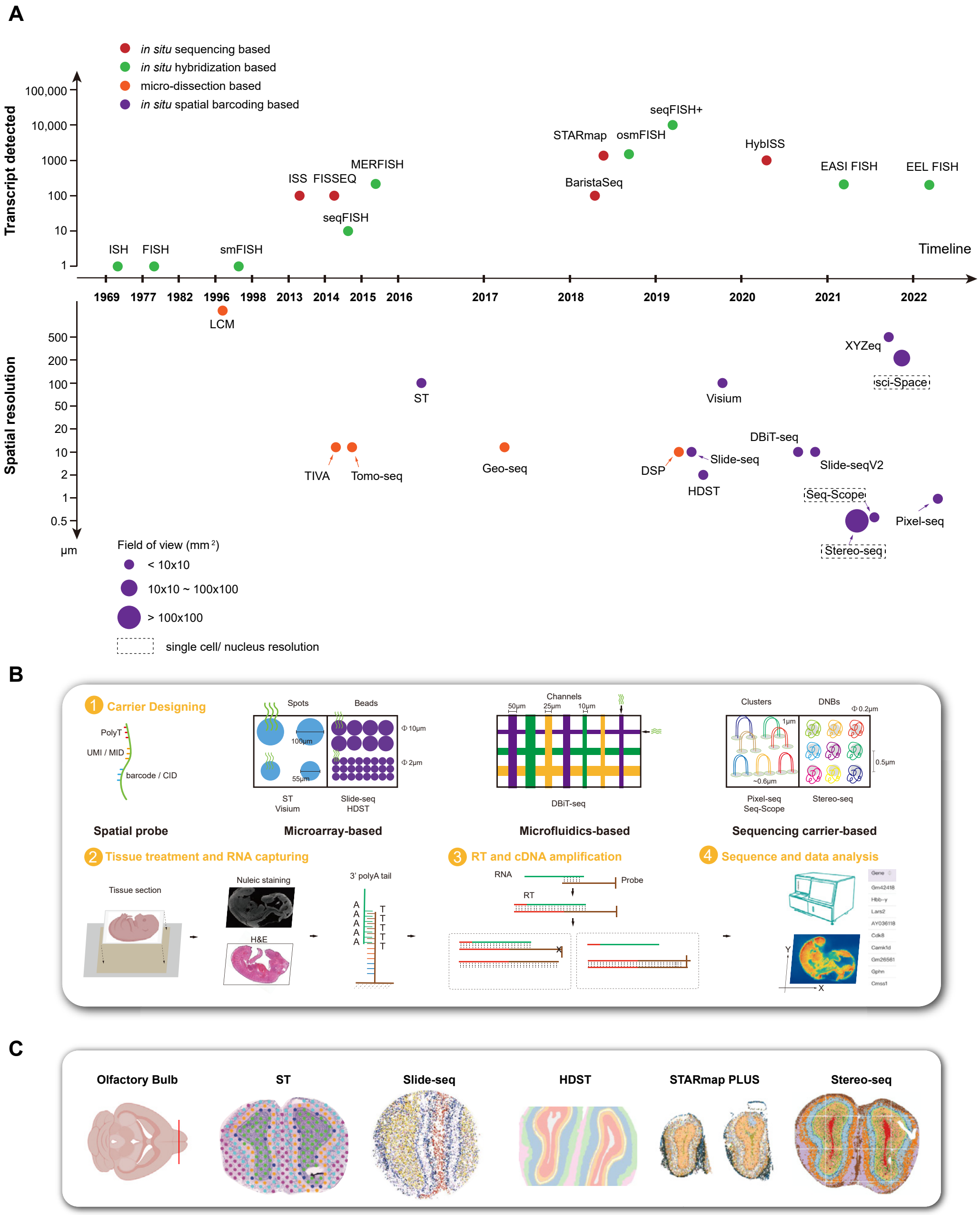

Fig. 1. A brief timeline and methodology of landmark SRT methods mentioned in this review. **A**: Timeline of the major SRT technologies. Approaches are presented in distinct colors according to their categories (red, *in situ* sequencing based method; green, *in situ* hybridization based method; orange, micro-dissection based method; purple, *in situ* spatial barcoding based method). The size of the purple circle donates the field of view of each method roughly. Dashed box represents the single-cell/nucleus resolution. The *x*-axes represent the timeline. The *y*-axes represent the number of transcripts detected (upper) and spatial resolution (bottom) in their experiments, respectively. **B**: Workflow of *in situ* spatial barcoding based method. Step 1, strategies for the carrier designing. Step 2, the process of tissue section indexing and RNA capturing. Step 3, strategies of product releasing. Step 4, sequencing and data analysis. **C**: Applications of different SRT technologies in mouse olfactory bulb. From left to right: ST (image adapted from Ståhl et al., 2016), Slide-seq (image adapted from Rodriques et al., 2019), HDST (image adapted from Vickovic et al., 2019), STARmap PLUS (image adapted from Shi et al., 2022), and Stereo-seq (image adapted from Chen et al., 2022a). SRT, spatially resolved transcriptomics; ST, spatial transcriptomics; HDST, high-definition spatial transcriptomics.

sequential cycles of hybridization. In terms of scalability, MERFISH and seqFISH+ could each provide a pertinent analysis at the level of hundreds-to-thousands and roughly ten thousand transcripts, whereas seqFISH could simultaneously determine RNA transcripts at the level of dozens. There are also additional techniques like enhanced electric FISH (EEL FISH) (Borm et al., 2023) and expansion-assisted iterative fluorescence FISH (EASI FISH) (Wang et al., 2021); of note, the latter approaches aim to address the fact that seqFISH and MERFISH only allow the profiling of a limited tissue size while, in parallel, being time-laborious.

Fast-moving of sequencing-based spatially resolved transcriptomics

As sequencing technology progressed, the sequencing-based SRT technologies have come into reality, with the obvious preponderance that no prior knowledge is required and that the target molecule can be captured without bias. Although there are still many challenges, the spatial resolution and capture efficiency are evolving very fast in this field. So far, the sequencing-based SRT technologies include (a) micro-dissection-based method (Junker et al., 2014; Chen et al., 2017), including laser capture microscopy coupled with Smart-seq2 (LCM-Seq) (Nichterwitz et al., 2016) and geographical position sequencing (GEO-Seq) (Chen et al., 2017; Cui et al., 2019; Peng et al., 2019); (b) *in situ* sequencing-based method (Ke et al., 2013; Lee et al., 2014; Chen et al., 2018; Wang et al., 2018b), e.g., STARmap, HybISS, and STAPmap PLUS; (c) *in situ* spatial barcoding based method (Ståhl et al., 2016; Rodriques et al., 2019; Vickovic et al., 2019; Chen et al., 2022a). The micro-dissection-based method uses different ways such as cryosection, micro-dissection, and UV-release to obtain the target area of interest. This method can achieve transcriptome-wide detection; however, due to the limitations of the release method, reaching high resolution remains difficult. *In situ* sequencing-based method amplifies the target signal *in situ* using padlock or rolling circle amplification followed by identification of base signal with a microscope. Similarly, for the imaging-based SRT technologies, these methods achieve subcellular resolution; however, given the limitation of the requirement of prior knowledge and limited cell space, this method could only capture targeted transcripts with restricted throughput. *In situ* spatial barcoding based method is the most recent breakthrough, which uses high-density barcoded probes fixed on a carrier to capture spatial RNA inside tissue sections, which is a process characterized by unbiased capture of whole transcripts in tissues.

The workflow and core aspects of *in situ* spatial barcoding based method

Considering the core principles of SRT technologies, the workflow of *in situ* spatial barcoding-based method could summarize into four basic parts: (a) '*Carrier*' designing. The carrier linked with spatial probes is the basic region to realize *in situ* RNA capturing, which is also the key part to greatly improve the performance of the technologies, such as the density changes of unique spatial probes from a lower level in glass slide surfaces to a higher level in sequencing chip surfaces determine the spatial resolution from multiple cells to a single-cell or subcellular level. (b) Tissue treatment and RNA capturing. Fresh-frozen tissues and formalin-fixed, paraffin-embedded (FFPE) tissues are the most common subjects currently in the tissue treatment step for SRT. Fresh-frozen tissues have higher RNA quality compared to FFPE and retain more comprehensive nucleic acid information. Of note, these facts explain why fresh-frozen treatment is the most common method for SRT currently, and FFPE is the better choice for tissues that need to be preserved for a long time and need to maintain tissue morphology. The tissues need to be sliced as thinly as about 10 μm, and the tissue sections are placed on the surface of the carrier to perform permeabilization using protein enzymes; these treatments could enable the spatial probes to efficiently capture RNA in tissues. (c) Reverse transcription (RT) and complementary DNA (cDNA) amplification. The captured RNA and spatial probes are linked to create a long chain to perform RT and generate cDNA with spatial barcodes. Some SRT technologies utilize cDNA as a template to synthesize the second chain and then release the second strand to amplify. Conversely, others use enzyme or chemical regent to straightly release cDNA from carriers to amplify. (d) Library construction, sequencing, and data analysis (Fig. 1B).

Therefore, the major technical challenges of such methods include (a) how to construct a special carrier innovatively called a '*capturing area*' that could effectively deliver the spatial barcode with polyT probes; (b) the number of spatial barcode probes in each capturing spot that tremendously influences the efficiency of capturing mRNA; (c) the spatial resolution for existing SRT technologies, which is determined by the width between nearby capturing spots (center to center) and the diameter of each spot (spot size); (d) capture area/throughput, which mainly depends on the number of capturing spots considering that any specific technology presents a certain (not unlimited) degree of resolution. Considering the above four key technological aspects in SRT, we will attempt to highlight below some of the latest *in situ* spatial barcoding-based methods from low to high spatial resolution. In that way, we will provide a comprehensive understanding of technical features and give guidance for researchers to choose the appropriate method for different contents (Table 2).

Microarray-based *in situ* spatial barcoding technologies

Utilizing glass surfaces or beads to link the spatial probes to generate capturing carriers represents a breakthrough for SRT technologies in the early stage. Moreover, by enhancing the density of unique spatial probes in glass surfaces or decreasing the diameter of beads, the spatial resolution would be improved to higher precision. The Spatial Transcriptomics (ST) (Ståhl et al., 2016) is the first *in situ* spatial barcoding-based technique, which utilized the glass slide surface to immobilize the spatial barcodes and polyT probes to generate thousands of capturing spots to create the capturing area. The spatial probes in ST are consistent inside each capturing spot but variable between two spots; hence, the minimum distance between capturing points determines the spatial resolution. The center-to-center distance between two adjacent capturing spots is 200 μm, which represents the spatial resolution. Likewise, the diameter of capturing spots is 100 μm, which means that the mRNA inside more than one cell could be captured simultaneously. In 2019, based on ST, Visium was developed by 10× Genomics with a higher spatial resolution of 100 μm (center-to-center distance between two adjacent capturing sports), and the diameter is 55 μm, which is smaller than that in ST but still bigger than the diameter of most cells. Thus, how to further enhance the spatial resolution still remains a continuous challenge in ST/Visium.

Considering the overall strategy for generating the capturing area, the Slide-seq/Slide-seqV2 and High-Definition Spatial Transcriptomics (HDST) are quite similar. Both of them utilize the beads with the pre-synthesized spatial barcodes and polyT probes to lay on a flat surface to form the capturing area, instead of directly printing the probes onto glass slides in ST. In both Slide-seq/Slide-seqV2 and HDST, each bead contains a special barcode sequence that serves as the basic unit of spatial resolution which is determined by the center-to-center distances between adjacent beads. In Slide-seq/Slide-seqV2, the spatial resolution is 10 μm, while in HDST, it is 2 μm. If only considering the physical size of cells, such as 10 μm × 10 μm, Slide-seq/Slide-seqV2 and HDST are the first or have almost reached single-cell resolution in barcoding-based approaches. Nonetheless, the capturing efficiency of each barcoding bead is not satisfactory for utilizing the single-cell region as the minimum analysis unit, which needs enough captured RNA information to perform statistically significant calculations. Therefore, improving the efficiency of RNA capture will be a crucial development factor for such high-spatial-resolution SRT approaches.

Microfluidics-based *in situ* spatial barcoding technologies

Integrating microfluidics with SRT technologies is a compatible strategy, which could adjust the size of the channel to change the spatial resolution, while free spatial probes could make spatial multi-omics more easily implemented. However, considering the limitations of engineering materials, the spatial resolution of this method would be hard to achieve the single-cell resolution. Deterministic barcoding in tissue for spatial '-omics' sequencing (DBiT-seq) is the first microfluidics-based technology, which simultaneously measures mRNA and protein spatial information via NGS. The capturing area with unique spatial barcodes and polyT probes is not entirely pre-synthesized in DBiT-seq, unlike the above technologies. Two polydimethylsiloxane (PDMS) microfluidic chips with parallel

Table 2
The main technical parameters of *in situ* spatial barcoding based methods.

Name	Resolution	Capture efficiency	Distance of spots	Diameter	Capturing area	Tissue thickness
ST	Multiple cells	3188 UMIs, 1485 genes/ 100 μm × 100 μm (mouse olfactory bulb)	200 μm	100 μm	6.5 mm × 6.8 mm	10 μm
SM-Omics	Multiple cells	11,261 UMIs, 3748 genes/ 100 μm × 100 μm (mouse olfactory bulb)	200 μm	100 μm	~42 mm^2	10 μm
Visium	Multiple cells	15,377 UMIs/55 μm × 55 μm (mouse olfactory bulb)	100 μm	55 μm	6.5 mm × 6.5 mm	10 μm
Ex-ST	Multiple cells	~15,000 UMIs/55 μm × 55 μm (mouse olfactory bulb)	100 μm	55 μm (to 20 μm)	6.5 mm × 6.5 mm	15 μm
STRS	Multiple cells	262–1095 viral UMIs/55 μm × 55 μm (mouse heart infected with REOV)	100 μm	55 μm	6.5 mm × 6.5 mm	10 μm
SiT	Multiple cells	1917 UMIs, 974 distinct isoforms/ 55 μm × 55 μm (mouse olfactory bulb)	100 μm	55 μm	6.5 mm × 6.5 mm	10 μm
Matrix-seq	Multiple cells	5085 UMIs, 2611 genes/50 μm × 50 μm (mouse olfactory bulb)	20, 40, 100 μm	10, 20 or 50 μm	75 mm × 25 mm	10 μm
CBSST-Seq	Multiple cells	7133 UMIs, 3190 genes/50 μm × 50 μm (mouse brain)	100 μm	50 μm	–	10 μm
Spatial-ATAC-RNA-seq	Multiple cells	~12,000 UMIs, ~4000 genes/ 50 μm × 50 μm (mouse embryo)	100 μm	50 μm	–	10 μm
xDbit	Multiple cells	5000–20,000 UMIs, 1000–5000 genes/ 50 μm × 50 μm (mouse organs)	100 μm	50 μm	1.17 cm^2	10 μm
DBiT-seq (FFPE)	Multiple cells	520 UMIs and 355 genes/25 μm × 25 μm (mouse embryo)	50 μm	25 μm	2.5 mm × 2.5 mm	5 μm–7 μm
spatial-CITE-seq	Multiple cells	1972 UMIs, 1166 genes/25 μm × 25 μm (mouse spleen)	50 μm	25 μm	2 mm × 2 mm	10 μm
HDST	Nearly single cell	12 UMIs/10 μm × 10 μm (mouse olfactory bulb)	3 μm	2 μm	5.7 mm × 2.4 mm	10 μm
Slide-seq	Nearly single-cell	59 UMIs/10 μm × 10 μm (E12.5 mouse embryos)	10 μm	10 μm	Φ3.0 mm	10 μm
Slide-seqV2	Nearly single-cell	550 UMIs/10 μm × 10 μm (E12.5 mouse embryos)	10 μm	10 μm	Φ3.0 mm	10 μm
DBiT-seq	Nearly single-cell	2068 genes/10 μm × 10 μm (mouse embryos)	20, 50, 100 μm	10, 20, 50 μm	1 mm × 1 mm	7 μm
XYZeq	Single-cell	1009 UMIs, 456 genes per cell (mouse liver/tumor tissue)	500 μm	500 μm	–	25 μm
sci-Space	Single-cell	2514 UMIs, 1231 genes per cell (E14.0 mouse embryos)	222 μm	73.2 μm	18 mm × 18 mm	–
MASC-seq	Single-cell	27,427 UMIs, 6293 genes per cell (human adenocarcinoma cell line)	200 μm	100 μm	6.5 mm × 6.8 mm	Single-cell
Pixel-seq	Subcellular	977 UMIs/10 μm × 10 μm (mouse olfactory bulb)	~1 μm	~1 μm	75 mm × 25 mm or Φ40 mm	10 μm
Seq-Scope	Subcellular	~1000 UMIs/10 μm × 10 μm (mouse liver)	~0.6 μm	<1 μm	0.8 mm × 1 mm	10 μm
STOmics-GenX	Subcellular	~2.1-fold gene numbers compared to Stereo-seq (STOmics)	0.5 μm	0.22 μm	10 mm × 10 mm	8 μm
Stereo-seq	Subcellular	1450 UMIs/10 μm × 10 μm (mouse olfactory bulb)	0.5 μm	0.22 μm	Up to 132 mm × 132 mm	10 μm

channels deliver uncombined polyT-tagged barcodes (A1–A50, B1–B50) into formaldehyde-fixed tissue sections. Also, polyT probes combine with mRNAs, which means the mRNAs in regional channels are linked with barcodes (A/N, B/N). Then mRNA-cDNA complexes are synthesized in RT, and subsequent processes include the followings: digesting tissues, collecting mRNA-cDNAs, amplifying templates, and generating libraries for NGS. The pixel size of channels, which determines the spatial resolution, contains 10 μm, 25 μm, and 50 μm. Even though the spatial resolution of DBiT-seq is nearly close to the single-cell level, the capturing and spot analysis are still not equivalent to the single cell; rather, they are relatively comparable to other high-resolution technologies, such as Slide-seq/Slide-seqV2 and HDST.

Sequencing carrier-based *in situ* spatial barcoding technologies

The combination of NGS '*chips*' and SRT technologies is of key importance to perform spatial analyses at single-cell resolution, which requires two rounds of sequencing. The carrier could be sequencing chips or similar chips, which could carry a higher density of spatial probes distribution. Also, the sequence of spatial probes is not required at the beginning of linking the spatial probes on the chip surface, but it could be sequenced during the linking process or at the first-round sequencing. Then put the tissue sections on the surface of the chips to capture RNA and finally do the second-round of sequencing which needs to be aligned with the first-round sequencing data to obtain the spatial location of each RNA, and which enables to achieve a spatial resolution at the single-cell level.

Polony-indexed library-sequencing (Pixel-seq) utilizes a 'stamp gel' as templates with ~1 μm clonal DNA clusters including polyT probes and spatial barcoded, which could be copied to many 'copy gels' to perform the capturing areas with 3′ polyT probes on a glass slide. The distance of adjacent and special DNA clusters that determine the spatial resolution of Pixel-seq is ~1 μm, combined with the segmentation of a weighted network and the transcripts' segregation into cell masks to realize the spatial single-cell resolution. Considering that the capturing cell layer is the only one which is adjacent to copy gel surface instead of multiple cell layers depending on the permeabilization in other *in situ* barcoding based methods, the capturing efficiency in Pixel-seq is nearly similar to the SRT methods with permeabilization, such as a mean of ~1000 unique molecular identifiers (UMIs) (10 × 10 mm^2 in mouse tissue). However, that still needs to be enhanced to higher captured UMIs in one cell region to achieve more biological identifications. Due to the gel-to-gel copied strategy, many '*copy gels*' could be generated from one 'stamp gel' template with the same spatial barcode probes, even the 'copy gel' could serve as the '*stamp gels*' for subsequent fabrication rounds, so that only one or a few '*copy gels*' need to be sequenced to obtain the spatial barcode information. Instead, every capturing chip of Seq-Scope and Stereo-seq needs to be sequenced separately, for which Pixel-seq strategy would reduce the fabrication cost and time. Seq-Scope, based on a solid-phase amplification, provides a ~0.5 μm −0.8 μm (~0.6 μm on average) spatial resolution beyond the previous *in situ* spatial barcoding-based technology, published in 2021. The library structure contains High-Definition Multimedia Interface (HDMI) (the spatial barcodes), polyT probes, and other primers, with the library being amplified to clusters with the PCR adapters in the cell surface. After the first round of sequencing, which aims to identify the unique spatial barcodes in the regional physical array, the polyT probes domain is then exposed to generate HDMI-encoded RNA-capturing array (capturing area) for tissue sections. Similar to the ST and HDST, Seq-Scope enables the tissue section sequencing and H&E staining; however, in Seq-Scope, the H&E staining image can be segmented based on the cell boundaries. The image and sequencing data can then be merged for further analysis based on the real single cell as the minimum spot.

The recently published Spatial Enhanced Resolution Omics-sequencing (Stereo-seq) technology is the first method that has achieved both spatial subcellular resolution and centimeter scale field-of-view, using DNA nanoball (DNB) on sequencing chips as the spatial barcoding strategy. Based on the 220 nm (diameter) DNB-patterned array on the lithography chip, Stereo-seq overcomes the limitation of the physical distance between spots to achieve the nanoscale spatial resolution or super-resolution. This Stereo-seq chip combines the high resolution of DNB with the high probe density per unit area, enabling efficient capturing of the spatial transcriptome at the genome-wide level with nanoscale resolution. Moreover, depending on the track-line features brought by DNB arrays, which are stained with nucleic acid dye and photographed together with tissue section after fixation and before permeabilization reaction, Stereo-seq can accurately record the histological cell morphology with mRNA transcripts detected from sequencing, thus enabling further single-cell segmentation. Meanwhile, compared to image-based methods, Stereo-seq has a greater but still suitable diffusion distance; for example, regarding the average diffusion of one measured gene, Stereo-seq performed a diffusion level of 6.84 μm compared to smFISH which performed a level of 5.32 μm. Besides, Stereo-seq also reports its capability of larger field-of-view ST by using 5 × 3 cm^2 chips for macaque brain tissue (Chen et al., 2022b). Overall, with the benefit of the high-density barcode carrier (with considerable probes and nanoscale resolution) and track-line features by the DNB chips, together with the high molecular capture efficiency and suitable RNA diffusion, Stereo-seq enables to decipher tissue complexity with the spatially resolved single-cell level and centimeter field-of-view.

Trends in the methods of spatially resolved transcriptomics

Despite the remarkable progress in spatially resolved transcriptome techniques over the past few years, the following aspects still need to be improved: (a) mRNA capturing efficiency is still a major challenge. Although Stereo-seq shows a significantly higher number of detected genes, for example, at single-cell size area, 1450 UMIs (mouse olfactory bulb, 100 μm^2) are detected in Stereo-seq, 848 UMIs (mouse liver, 100 μm^2) in Seq-Scope, and 550 UMIs (mouse embryo, ~100 μm^2) in Slide-seqV2 (Cho et al., 2021; Stickels et al., 2021; Chen et al., 2022a). Compared to the sensitivity of the current single-cell RNA sequencing (scRNA-seq) technologies, for example, ~40,000 UMIs (~6000 genes) can be detected in 10× Chromium and DNBelab C4, ~8000 genes in Smartseq2 (Liu et al., 2019; Ding et al., 2020), significant improvement is still needed to enhance the performance of cell classification in SRT technologies. Currently, SRTs propose a new strategy that uses poly (A) polymerase to add poly (A) tails to RNAs *in situ* to enable the detection of the full spectrum of RNAs, simultaneously, capturing the adding-poly (A) RNAs that used barcoding based methods. The latter could enhance the capturing efficiency compared with only capturing the mRNA with native poly (A) tails, which could not be measured after degradation. This issue, RNA degradation, is a big challenge in some clinical tissues or easy-degradated samples (McKellar et al., 2022).

(b) The spatial resolution of *in situ* spatial barcoding based methods has an enhanced trend: from the several mixture cells per capturing spots in ST/Visium to the nearly single-cell resolution in Slide-seq/Slide-seqV2, HDST, and DBiT-seq, even to the true single-cell resolution based on the cell segmentation in Seq-Scope and Stereo-seq; however, further developing the resolution is still necessary for new insights into spatial gene regulation and biology inside cells. The expending microscope (ExM) is a potential technology to enhance the spatial resolution in SRT technologies with the swellable gel material, which has been used in Ex-Seq (Alon et al., 2021) and STARMap/STARMap PLUS (Wang et al., 2018b; Zeng et al., 2023) belonging to the *in situ* sequencing-based method. For *in situ* barcoding based methods, expansion spatial transcriptomics combined the ExM and Visium to push the resolution of Visium array from 55 μm to 20 μm and to also improve RNA capturing efficiency per region (Fan et al., 2022). Moreover, if the higher-spatial-resolution SRT techniques, such as Seq-Scope and Stereo-seq, are combined with the expansion microscope, the spatial resolution would be further enhanced from ~500 nm to ~100 nm or even less than 100 nm, which may enable the new insights into biology.

(c) Universality of sample types. Most of the current ST methods are applicable to fresh-frozen tissues, which are not conducive to long-term preservation, and they are always prone to deformation and to gene diffusion during the step of tissue permeabilization. Some techniques have been implemented in other samples such as FFPE tissues, such as Visium (Gracia Villacampa et al., 2021) and Xenium In Situ Technology in 10× genomics and DBiT-seq (Liu et al., 2020b). However, considering the inadequate quality of RNAs in FFPE samples, the performance of the SRT dataset, and the number of captured genes, much space is left for further development. Besides, paraformaldehyde embedding is also a common way of tissue preservation, which may help in the preservation of tissue morphology and *in situ* RNA capture with higher accuracy and less diffusion. Of note, only a few SRT technologies are compatible with the paraformaldehyde solution and are currently available, such as Visium (Gracia Villacampa et al., 2021).

(d) It is important to obtain spatial multi-omics which may include gene expression, hematoxylin and eosin (HE) staining, and protein distribution within the same tissue section to fully comprehend the gene expression profiling of various tissue regions with varying states. Some researchers have already achieved analysis of spatial transcriptome with HE staining, which can provide histological region and cytomorphology. Of course, it would be more promising to achieve HE staining, immunofluorescence, spatial target protein, or spatial epigenome analysis on the same slice with the spatial transcriptome. Now, SRT technology tends to be mature and stable, which provides guidance for the research and development of spatial multi-omics technology. Currently, several spatial multi-omics technologies with good compatibility have been introduced, such as SM-Omics (a platform for spatial transcriptome and proteome based on ST) (Vickovic et al., 2022), Slide-DNA-seq (a spatial genome technology developed based on Slide-seq) (Zhao et al., 2022), Spatial-CUT&Tag (Deng et al., 2022a) and spatial-ATAC-seq (Deng et al., 2022b) based on the DBiT-seq platform, spatial proteome technology, and proExM (ProteomEx) based on the ExM (Wassie et al., 2019). However, there is still much space for improvement in the spatial resolution, detection sensitivity, and protein detection quantity of existing technologies. Meanwhile, the combination of SRT technologies with other traditional molecular biology techniques is also an important area for future development. SRT technology can answer questions about the molecular biological characteristics and heterogeneity of cells or nucleic acids at the spatial level, but the detection and interpretation of biological information at a certain biological stage or a certain biological level are limited. Therefore, more abundant biological information can be obtained by combining with other technologies to realize detection during cell differentiation, gene editing, transcription event or local targeting of RNA. Now, some powerful combination techniques have been reported, such as Space-TREX (Holgersen et al., 2021), which combines ST and linage tracing techniques to detect the variation in cell types and spatial location during progenitor cell differentiation; Perturb-map (Dhainaut et al., 2022), which combines CRISPR technology with ST technology to identify regulators of the tumor microenvironment; slide-TCR-seq (Liu et al., 2022b), which compares T cell reporters and transcriptomes to show different immune niches and interactions in the adaptive immune response; RIBOmap (Zeng et al., 2022) based on STARmap (Wang et al., 2018b), which can detect the spatial state of protein synthesis after RNA translation *in situ*; and TEMPOmap (Ren et al., 2022), which can detect the spatial and temporal evolution of new transcriptome with spatiotemporal resolution *in situ*. In the future, it is expected that more joint technologies will emerge in various research areas, such as spatial neuronal projection detection using viral transfection, combined use of cytogenetics and spatial transcriptome techniques, and among others.

Tools and challenges in spatially resolved transcriptomics data analysis

Compared to single-cell '-omics', the high-dimensional spatially resolved omics brings a huge amount of data and, therefore, big challenges in data analysis. The overall process of spatial dataset analysis consists of several parts: (1) spatial matrix generation, of which the core function is to generate a gene expression matrix with spatial location information from sequencing data according to barcoding schemes of different technologies; (2) image registration and cell segmentation, which is the key step to achieve spatial single-cell analysis; (3) downstream analysis, which includes (a) data preprocessing, such as data quality control, normalization, and batch correction; (b) gene imputation; (c) spatial domain identification and high-variable gene detection, mainly in single-cell resolution and spatial context; (d) clustering, integrating gene expression, spatial information and/or image information; (e) cell type annotation, containing using methods for scRNA-seq and deconvolution to single-cell resolution (for lower-resolution SRT); (f) cell communication, containing using methods for scRNA-seq and connecting to the spatial arrangement; (g) trajectory analysis; (h) alignment and integration of adjacent-slide SRT dataset and further 3D reconstruction; (i) integration ST with other modalities such as ATAC or protein; (4) database construction, of which aim is to story the high-quality SRT data, establish an interactive user platform and a unified international standard for SRT data. Below, we summarize the tools and challenges of the key aspects in high-resolution SRT analysis.

Spatial matrix generation

The wide matrix form of the so-called 'spatial expression matrix' is a 2D non-negative matrix, where the row names are genes and the column names are pairs of 2D XY coordinates. To reduce storage space, the spatial expression matrix can be processed through a long matrix, which only contains the 3D coordinates (X, Y, gene) and the positions of non-zero values. In both settings, obtaining the spatial expression matrix is the first step toward upstream analysis of ST, where the critical aspect lies in deciphering the spatial location of each mRNA transcript.

In imaging-based SRT, the 2D fluorescence signals, which are generated during each round of sequencing, are directly read on the imaging platform, which generates the fluorescence signal and its relative position in the image. The challenge in this process lies in the need for adequate resolution of the fluorescence signal in order to distinguish crowded molecules by harnessing sophisticated microscopy techniques, such as expansion microscopy (Chen et al., 2015a) and/or advanced bioinformatics algorithms (Xia et al., 2019). By using lookup tables and signal calibration, the fluorescence signal can be correlated with genes. After obtaining the fluorescence signal representing the molecule and its spatial coordinates, the spatial expression matrix can be derived. Then, by applying cell segmentation algorithms, the boundary region of each cell can be delineated, thus mapping the spatial coordinates to single-cell spatial expression matrix (further detail are explained in the following section).

In sequencing-based SRT, different methods exist for obtaining the spatial expression matrix. Similarly to standard single-cell sequencing techniques, there are two libraries in the sequencing-based SRT: read 1 containing the barcode and UMI sequences and read 2 containing cDNA sequences. However, the difference lies in the fact that in spatial techniques, the barcode is associated with spatial coordinates, not cells. Of note, currently, each technique has its own way of associating barcodes with spatial coordinates: (a) in DBiT-seq, ST, and other techniques, the spatial location of the barcode is predetermined, and each barcode has a previously known spatial coordinate (Ståhl et al., 2016; Liu et al., 2020c); (b) in seq-Scope technology, sequencing-by-synthesis technology ensures that the corresponding spatial coordinates can be obtained during the synthesis process of the spatial chip HDMI (Cho et al., 2021); (c) in Stereo-seq, after loading the DNB into the spatial chip, the first round of *in situ* sequencing has been performed to obtain the barcode sequences and corresponding spatial coordinates for each DNB, followed by spatial transcriptome experiment and capturing of cDNA sequences in the second sequencing round (Chen et al., 2022a). After establishing the correspondence between barcodes and coordinates in read 1, the alignment of read 2 to the genome enables the determination of the spatial position of gene expression (Navarro et al., 2017). Overall, this workflow is consistent with the methodology employed in single-cell transcriptome analysis.

Image registration and cell segmentation

The high-resolution SRT dataset usually contains immunohistochemical, hematoxylin-eosin staining (H&E), and/or nucleic acid staining images to obtain subsequent cell/nuclear segmentation and to complement the biological features at the cellular level (Vickovic et al., 2019; Chen et al., 2022a). Thereby, the first step is the alignment of stained images with the expression matrix, which forms the basis for integrating macroscopic cellular states and genetic molecular information for analysis. To this end, image registration tools, such as Spot Detector, provide an automatic process of image and spots positions alignment, mainly including the step of alignment, framing, and verification (Wong et al., 2018). However, with the emergence of higher-resolution SRT technologies, a more accurate alignment process is required. For example, in the registration process of Stereo-seq, the matrix of each DNB spot with expression is converted to a grayscale image and then performed alignment with the nucleic acid staining image manually using the track-line features (Chen et al., 2022a; Cheng et al., 2022). The next step is to obtain spatial single-cell/nuclear data after segmentation based on the cell/nuclear location of the stained images, which provides a bridge to achieve spatial single-cell resolution. Based on the various considerations to achieve cell segmentation, multidimensional mRNA density estimation (SSAM) calculates cell types based on RNA distribution without segmenting cell boundaries (Wong et al., 2018). In contrast, Baysor enables cell boundary identification and segmentation annotation (Petukhov et al., 2022). Joint cell segmentation and cell type annotation (JSTA) uses scRNA-seq-defined cell types to assist in the identification of spatial data (Littman et al., 2021).

Integration of SRT and single-cell dataset

Towards the aim of achieving spatial single-cell analysis in lower-resolution SRT technologies, deconvolution from the scRNA-seq dataset is commonly required instead of direct cell segmentation. One of the commonly used solutions is to integrate SRT with the scRNA-seq datasets. Of note, one way that two datasets can currently be integrated is through probabilistic models. The expression of individual cells in single-cell data can often be modeled as negative binomial distribution, Poisson distribution, and so on. Robust cell type decomposition (RCTD) (Cable et al., 2022), stereoscope, and cell2location use the above probabilistic model to integrate single cells and give the deconvolution cell types. SPOTlight (Elosua-Bayes et al., 2021) and SpatialDWLS (Dong and Yuan, 2021) use the topic similarity obtained after NMF dimension reduction to determine SRT cell types. CellTrek (Wei et al., 2022a) projects the SRT and scRNA-seq data into the same latent space and Multivariate RF is modeled. Tangram is a deep learning method, which calculates the corresponding relationship between SRT spots or cells after cell segmentation through a neural network (Biancalani et al., 2021). However, the strategy of integrating SRT and scRNA-seq dataset still presents some challenges, including the possible outcome of ignoring or losing some parts of information of rare-type cells due to current technological and algorithmic limitations. For example, in low-spatial-resolution SRT techniques, one capture spot simultaneously covers multiple cell regions and captures mixed RNAs of multiple cells, rendering it impossible to accurately distinguish the mRNA signal *in situ* of each cell and divide the boundary of RNA signal of a single cell, thus failing to conduct subsequent analysis with single-cell precision. At the same time, the transcriptomic features of rare cell types are not clearly distinguished from major cell types; however, the specific information of rare cell types retained during data integration could be a potential point to further identify the specific markers of rare cell types. For example, RaceID provides a strategy and shows good performance for identifying rare cell information from multiple cell populations (Grun et al., 2015; Herman et al., 2018).

Gene imputation

Gene imputation represents a crucial step to ameliorate the quality of SRT data, especially those from imaging-based SRT approaches. Due to the limit of barcode variation, imaging-based SRT usually only captures the genes of interest (e.g., seqFISH, MERFISH, STARmap), resulting in a narrower range of gene detection. To this end, several methods have been developed for gene imputation purposes, such as Tangram (Biancalani et al., 2021), gimVI (Lopez et al., 2019), SpaGE (Abdelaal et al., 2020), Seurat (Satija et al., 2015), and LIGER (Liu et al., 2020a). Using the combined analysis of scRNA-seq and imaging-based SRT, the unbiased gene values captured by the scRNA-seq can be applied to calculate the lost genes in imaging-based SRT. SpaGE, Seurat, gimVI, and Liger can be co-embed scRNA-seq and SRT into a common latent variable space to align the two technologies and perform gene imputation based on K-nearest neighbor or matching probability. Tangram learns the alignment of scRNA-seq and SRT using deep learning directly and imputes the missing gene counts by soft assignment.

Cell communication

The emergence of SRT technologies has enabled the measurement of cell–cell communications directly within specific microenvironments due to the analysis of cell interaction, considering that spatial distance is more conducive to the analysis of comparing short-range interactions (Liu et al., 2022c). As a result, the false positive results determined only by the amount of RNA expression are excluded. Based on the SRT dataset, the expression of ligand receptors and the determination of spatial distance exist simultaneously, so that the cells can be judged to interact with each other (Cang et al., 2022; Kim et al., 2022). Importantly, regarding the spatial aspect of the cell–cell interactions, localization of the ligand in the corresponding subcellular location, which standard single-cell RNA-Seq experiments are unable to decipher, is of paramount importance. To this end, physical interactions (PIC)-seq approaches provide a high-throughput solution to study the interactions due to spatial proximity, while advanced algorithms considering the constants of intercellular communication can provide additional input. Of note, several computational tools, such as CellChat (Jin et al., 2021) and CellCall (Zhang et al., 2021) have been developed in that regard.

To identify the intercellular communication using SRT dataset, different tools apply several methods; for instance, Graph Convolutional Neural networks for Genes (GCNG) analyzes intercellular gene interactions (Yuan and Bar-Joseph, 2020), SVCA quantifies the effect of intercellular interactions on gene expression (Arnol et al., 2019), and DIALOGUE interprets multicellular collaborative programs (Jerby-Arnon and Regev, 2022). Moreover, both NICHES (Raredon et al., 2023) and HoloNet (Li et al., 2022b) consider the ligand and receptor expression profiles, in order to visualize heterogeneous signaling archetypes, or to develop a graphic neural network model for exploring the intercellular communication-related transcript events. Last, the SpaTalk tool models and scores the ligand-receptor-target signaling network based on network biology approaches, such as knowledge graph approaches (Mohamed et al., 2021), in order to uncover cell–cell communications for SRT dataset at a universal scale (Shao et al., 2022).

Challenges in spatially resolved transcriptomics data analysis

Although several analytical tools have been published, as the cost of SRT technologies declines and the technology becomes widely available, more data analysis challenges will emerge. It may include the following aspects: (a) the volume of SRT data is huge. With increasing datasets with high density and large field of view, new storage methods and data preprocessing speed of SRT data will be urgent issues to be solved in the future; (b) batch effects often occur between different slices due to differences in experimental batches, operators, or operation methods. How to correct the batch differences correctly so that different slices can be compared with each other to obtain more accurate and reliable analysis conclusions remains a major challenge. Of note, several techniques have been developed to address batch effects regarding single-cell RNA-seq approaches (Berglund et al., 2020; Cheng et al., 2021; Dong and Zhang, 2022; Fang et al., 2022). Currently, de-batch methods for scRNA-seq, such as canonical correlation analysis (CCA), have already been applied to the spatial data. Nevertheless, all these approaches need to be standardized for SRT considering the complexity of the latter; (c) since the data of each slice in SRT cannot represent an entire organ, 3D reconstruction using SRT data will become consistently important in the future, as elegantly shown by a recent study on "3D-cardiomics" for heart tissue (Mohenska et al., 2022). Current advances in the 3D field are mainly based on the 3D reconstruction of multiple adjacent slices of the same tissue, which rely on both the stability of experimental techniques, which allow for integrated analysis of multiple adjacent slices, and analytical tools, which are capable of 3D reconstruction, such as probabilistic alignment of ST experiments (PASTE) (Zeira et al., 2022). Nonetheless, more efficient analytical methods are still needed in the future to address many of the current challenges facing 3D reconstruction, such as cell mapping and linkage reconstruction between adjacent slices. Besides, with the rapid progress of advanced analytical methodologies, these methods require systematic evaluations to determine the reliability and to perform guidance of the selection of analytical tools. Many benchmarking studies focusing on single-cell resolution and the choice of spatially variable genes shall be conducted, while other novel algorithms also need to be established including the standardization evaluation system, the accuracy of cell segmentation, the spatial domain identification, and the spatial clustering. Until now, the existing benchmarking studies so far have shown that some tools perform better in assessing how RNA transcripts are spatially distributed, while others have shown better performance regarding cellular deconvolution approaches or in their clustering accuracy (Li et al., 2022a).

Overall, the challenges that need to be addressed include data registration across slices, eliminating batch effects, data normalization, and filling gaps of information between slices. Additional challenges refer to (a) integrating ST with other modalities. Given the breakthrough of the spatial multi-omics method, it is highly desirable to improve an integrated algorithm to understand spatially resolved data with tissue functions, and to appreciate how SRT data will be combined with those derived from other methods, such as proteomics and chromatin accessibility (Kriebel and Welch, 2022); (b) how to identify spatial domains characterized by coherency in the gene expression patterns and histological features (Hu et al., 2021); (c) how to combine SRT data with several (and not singular) neighboring tissues processed through different slices (Zeira et al., 2022).

Spatially resolved transcriptomics in dissecting tissue complexity

Thanks to the remarkable progress in life science research, SRT has been widely applied in many fields for constructing gene expression landscapes in several tissues across multiple species at the spatial resolution level (Fig. 2; Table 3). One of the attracting fields is to reconstruct spatiotemporal transcriptomics atlas of embryo development at the millimeter-to-centimeter scale. For example, by using Stereo-seq, researchers obtained the spatiotemporal gene expression atlases of embryo development using the key and classical model animals including mice (Chen et al., 2022a), zebrafish (Liu et al., 2022a), and *Drosophila* (Wang et al., 2022). These studies firstly demonstrate that cellular resolution developmental trajectories and cell–cell interactions are both related to the developmental events at spatial level. In parallel to these studies, Wei et al. constructed a high-definition spatially resolved cell landscape of axolotl telencephalon using Stereo-seq (Wei et al., 2022b), uncovering new cell types participating in brain regeneration. By harnessing a combination analysis of ST, scRNA-seq, and *in situ* sequencing data, Asp et al. profiled the dynamics of gene expression during human cardiogenesis, identified the distributions and functions of various cell types in developing heart, and constructed a 3D organ transcriptomics-based map (Asp et al., 2019).

The second research direction is to 3D reconstruct organs at molecular and cellular resolution. For instance, a systematic molecular atlas of the whole adult mouse brain by employing unbiased ST mapping was first reported (Ortiz et al., 2020). By applying high-resolution and large field-of-view Stereo-seq technology, Chen et al. obtained a 3D spatially resolved cell type atlas for the cynomolgus monkey brain, with most of the sections achieving the area of 5×3 cm^2 (Chen et al., 2022b), enabling the study on cell

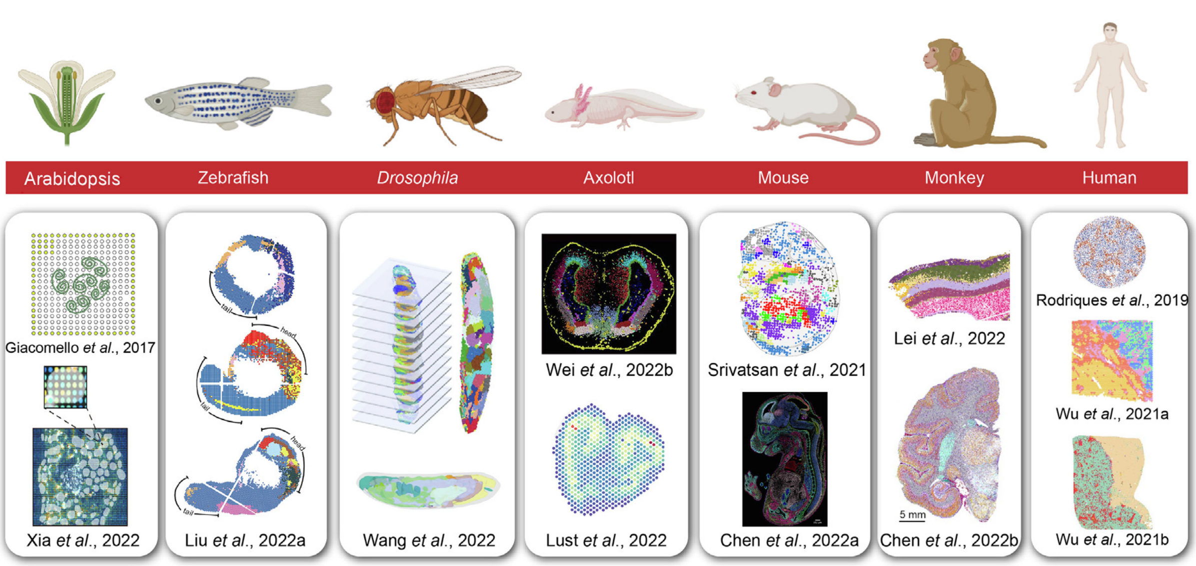

Fig. 2. Recent representative applications of spatially resolved transcriptomics. Applications of different SRT technologies in model organisms. Arabidopsis depicted through leaf bud which was analyzed by ST (upper) (image adapted from Giacomello et al., 2017) and Stereo-seq (bottom) (Xia et al., 2022). Zebrafish and *Drosophila* depicted through embryos by Stereo-seq (image adapted from Liu et al., 2022a and Wang et al., 2022). Axolotl depicted through telencephalon by Visium (upper) (image adapted from Lust et al., 2022) and Stereo-seq (bottom) (image adapted from Wei et al., 2022b). Mouse exampled depicted as embryos analyzed by sciSpace (upper) (image adapted from Srivatsan et al., 2021) and Stereo-seq (bottom) (image adapted from Chen et al., 2022a). Monkey depicted through its cerebral cortex (upper) (image adapted from Lei et al., 2022) and hemisphere (bottom) (image adapted from Chen et al., 2022b) by Stereo-seq. Human organism depicted through the liver organ analyzed by Slide-seq (upper) (image adapted from Rodriques et al., 2019), primary liver cancers by ST (middle) (image adapted from Wu, 2021a), and intrahepatic cholangiocarcinoma by Stereo-seq (bottom) (image adapted from Wu et al., 2021b).

Table 3
The key four-field articles of SRT technologies.

Field	Species	Tissue	Health state	Technology	References
Atlas	Mouse	Liver	Normal	smFISH	Halpern et al., 2017
	Human	Heart	Normal	ST	Asp et al., 2019
	Mouse	Brain	Normal	Visium	Joglekar et al., 2021
	Human	Dorsolateral prefrontal cortex	Normal	Visium	Maynard et al., 2021
	Human	Fetal liver	Normal	Visium, Stereo-seq	Gao et al., 2022
	Mouse	Intestine	Damage	Visium	Parigi et al., 2022
	Human	Lung	Normal	Visium	Kadur Lakshminarasimha Murthy et al., 2022
	Human	Human hematopoietic stem cell	Normal	Visium	Calvanese et al., 2022
	Arabidopsis	Leave	Normal	Stereo-seq	Xia et al., 2022
	Human	Immune system	Normal	Visium	Suo et al., 2022
Brain Atlas	Mouse	Hypothalamic Preoptic	Normal	MERFISH	Moffitt et al., 2018
Brain Atlas	Mouse	Ventromedial hypothalamus	Normal	seqFISH	Kim et al., 2019
Brain Atlas	Mouse	Brain	Normal	ST	Ortiz et al., 2020
Brain Atlas	Human, Mouse, Chicken	Cerebellar nuclei	Normal	STARmap	Kebschull et al., 2020
Brain Atlas	Mouse	Brain	Normal	HybISS	La Manno et al., 2021
Brain Atlas	Mouse	Motor cortex	Normal	MERFISH	Zhang et al., 2021
Brain Atlas	Mouse	Primary motor cortex	Normal	MERFISH	Booeshaghi et al., 2021
Brain Atlas	Mouse	Preoptic Neuron	Normal	MERFISH	Osterhout et al., 2022
Brain Atlas	Human, Mouse	Brain cortex	Normal	MERFISH	Fang et al., 2022
Brain Atlas	Mouse	Somatosensory cortex	Normal	MERFISH	Stogsdill et al., 2022
Brain Atlas	Macaque	Cortex	Normal	Stereo-seq	Lei et al., 2022
Brain Development	Mouse	Cerebral cortex	Normal	Slide-seqV2	Di Bella et al., 2021
Brain Regeneration	Axolotl	Brain	Injury-induced	Stereo-seq	Wei et al., 2022b
Brain Organoid	Human	Brain	Normal	Slide-seq	Uzquiano et al., 2022
Brain Disease	Human, Mouse	Spinal cord	Amyotrophic lateral sclerosis	ST	Maniatis et al., 2019
Brain Disease	Human, Mouse	Brain	Alzheimer's disease	ST	Chen et al., 2020
Brain Disease	Mouse	Brain	Alzheimer's disease	STARmap PLUS	Zeng et al., 2023
Brain Disease	Human	Brain	Melanoma brain metastasis	Slide-seqV2	Biermann et al., 2022
Development	Zebrafish	Embryo	Normal	tomo-seq	Junker et al., 2014
	Mouse	Embryo	Normal	Geo-seq	Peng et al., 2019
	Mouse	Gastruloids and Embryo	Normal	tomo-seq	van den Brink et al., 2020
	Human	Gastruloids and Embryo	Normal	tomo-seq	Moris et al., 2020
	Human	Intestine	Normal	Visium	Fawkner-Corbett et al., 2021
	Mouse	Embryo	Normal	seqFISH	Lohoff et al., 2022
	Mouse	Embryo	Normal	Stereo-seq	Chen et al., 2022a
	Human	Gonad	Normal	Visium	Garcia-Alonso et al., 2022
	Drosophila	Embryo and Larvae	Normal	Stereo-seq	Wang et al., 2022
	Zebrafish	Embryo	Normal	Stereo-seq	Li et al., 2022a
	Human	Fetal lung	Normal	Visium	He et al., 2022
Disease	Human	Pancreat	Pancreatic ductal adenocarcinomas	Visium	He et al., 2022
	Human	Skin	Squamous cell carcinoma	ST	Ji et al., 2020
	Human, Mouse	Liver	Hepatocellular Carcinoma	DSP	Sharma et al., 2020
	Human, Mouse	Lung	Acute respiratory distress	Visium	Boyd et al., 2020
	Human	Colorectal cancer	Colorectal cancer	DSP	Pelka et al., 2021
	Human	Glioblastoma	Glioblastoma	Visium	Ravi et al., 2022
	Human	Gastric Cancer	Gastric Cancer	DSP	Kumar et al., 2022
	Mouse	Skin	Injury-induced	Visium	Konieczny et al., 2022
	Mouse	Skin	Melanoma	Stereo-seq	Karras et al., 2022
	Human	Prostate, Skin, Lymph node, Glioblastoma multiforme, Breast	Malignant tissue	Visium	Erickson et al., 2022
	Human	Heart		Visium	Kuppe et al., 2022

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Table 3 (*continued*)

Field	Species	Tissue	Health state	Technology	References
			Myocardial infarction		
	Human	Cervical Squamous Cell Carcinoma	Cervical Squamous Cell Carcinoma	Stereo-seq	Ou et al., 2022
	Mouse	Lumbar spinal cords	Paralysis	Visium	Kathe et al., 2022
	Human	Oral cancer, Colorectal cancer	Oral cancer, Colorectal cancer	Visium, DSP	Galeano Niño et al., 2022

distribution heterogeneity and functions at whole-brain scale. Similarly, Lei et al. combined Stereo-seq with single-nuclei assay for transposase-accessible chromatin with high-throughput sequencing (snATAC-seq) and scRNA-seq to reveal differences in cell type-specific and regional-specific genes and regulatory elements of the primate cerebral cortex (Lei et al., 2022).

Besides, SRT technology is useful for the study of disease mechanisms. Compared with traditional pathological diagnoses, SRT also provides a very promising tool for future pathological studies. With the advancement of high-resolution SRT technologies, tumor boundaries, and microenvironments can be located and defined at the molecular level. For instance, to explore the heterogeneity and microenvironment of intrahepatic cholangiocarcinoma, Wu et al. first used SRT analyses, defined a 500 μm-wide zone centered bilaterally on the tumor boundary, and characterized the cellular and transcriptional heterogeneity in the tumor invasive front (Wu et al., 2021b). By researching colorectal adenocarcinoma using the SRT technology, Zhang et al. characterized the profiling of complicated intratumoral regions, and they identified areas of discontinuous inflammation. This study provided a broad picture of the intratumoral microenvironment, thus enabling the exploration of crosstalk among tumor, immune cells, and stromal cells (Zhang et al., 2022).

Another intriguing application of SRT technology is to study the cell landscape of plants at spatial resolution. The first spatial application of SRT strategies in plants was reported by Giacomello and colleagues, and it enabled accurate quantification of gene expression and pathway scores in different regions of Arabidopsis leaf bud (Giacomello et al., 2017). Recently, by using Stereo-seq, Xia et al. firstly distinguished *Arabidopsis thaliana* leaves' cell subtypes based on spatial information (Xia et al., 2022), indicating the power of high-resolution SRT in bringing new findings for plant biology.

Future perspectives

Despite the remarkable progress in SRT techniques over the past few years, some aspects which are still underway include the improvement of SRT-related technologies and bioinformatic algorithms (Fig. 3). Regarding the SRT technology, (a) for the analysis of spatial heterogeneity, the detection of poorly expressed genes, and

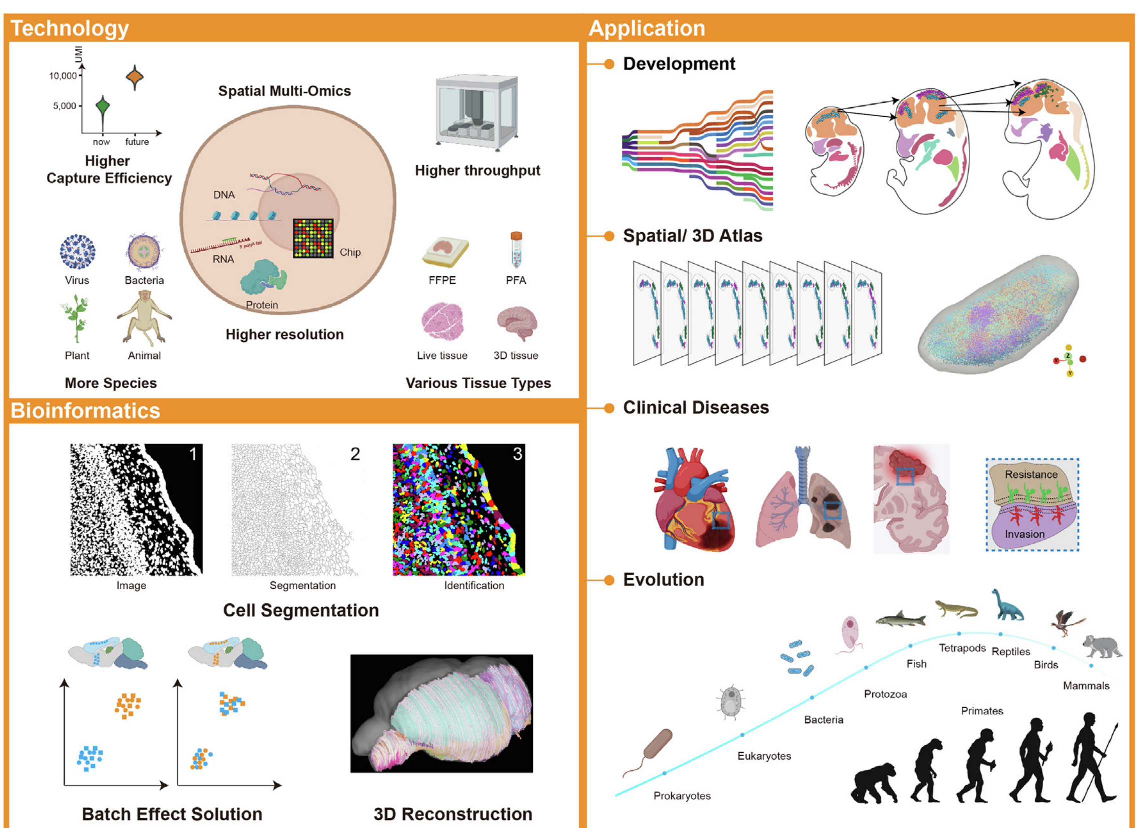

Fig. 3. Future perspectives of SRT-related technology, bioinformatics, and the fields of application. Continuous development of SRT technologies and bioinformatics algorithms accelerates the research in embryonic development, spatial atlas, clinical diseases, and evolution, providing the potential impact in both basic science and translational medicine.

the discovery of unreported maker genes, achieving higher spatial resolution and capturing efficiency reflects the future developmental paths; (b) realizing that full-length transcripts can capture the entire nucleic acid base sequence instead of only the 3′ tails of mRNA with polyA, they can work together with targeted gene capturing, which is focused on the few but specific genes; (c) in the long-term, sequencing-based SRT technologies will be combined with 3D tissue transparency technologies and *in vivo* imaging in the future to directly obtain gene expression patterns and cell spatial distribution characteristics from the 3D level; (d) in addition, spatial multi-omics, higher throughput work capability, and automated operation process are also major developmental directions of the SRT technology in the future. For bioinformatic analysis of SRT datasets, how to utilize the spatial information effectively to accurately reflect and encompass all features of biological phenomena is still a challenge that requires sophisticated tools to resolve. The latter is continuously being developed to resolve the data analytics issues, which emerged over the last years while decoding spatial information. Macroscopic life sciences events, such as species evolution, ontogeny, and disease development are, at their core elements, based on cell-scale fate transitions and molecular regulations of genetic elements. SRT technologies can integrate macroscopic representations with microscopic information to complement and/or provide new insights into existing studies or inspire new studies regarding spatial dimensions. In the future, as continuous optimization, updating iterations, and the cost decline of SRT technologies, it is foreseeable that SRT will become a fundamental tool to be fully applied to various fields such as 3D organ reconstruction, embryonic development and regeneration, evolution, and human disease, for both basic science and translational medicine.

Conflict of interest

The authors declare that they have no conflicts of interest.

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